How Economists Discuss Central Bank Digital Currencies

Themes, tone and stance, 2014–2026: a bibliometric and text-as-data study

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M2 – Private crypto

The paper in one slide

Research question. How has economists and institutional discourse on CBDCs evolved between 2014 and 2026, and to what extent does this evolution align with changes in the policy environment?

Who writes?	research economists, central banks and institutions
What do they write about?	themes and topics, and how they shift
How do they discuss CBDCs?	**tone and stance: the core of the paper**
Why, in their own words?	stated motivations
When, and with which environment?	bursts, policy events, sanctions

Scope. Descriptive: we measure changes and show what moves with them; we do not claim causes.

48,188 texts collected $\rightarrow$ 17,709 with CBDC as a theme $\rightarrow$ 4,710 written by economists

Outline

Introduction

Data

Who writes

What is written

How: tone and stance

Why: stated motivations

When: the environment

The discussion and its environment

Robustness

Conclusion

Introduction

Related literature: CBDCs

Related literature: what the field builds on

What this paper adds to 29 earlier reviews

Five recent bibliometric reviews:

Study	Data	Documents	Tools
Kvedaravičiūtė & Šapkauskienė (2024)	WoS + Scopus	228	VOSviewer
Fahad & Bulut (2024)	Scopus	169 of 323	Biblioshiny
Bapat & Sharma (2025)	Scopus	371	bibliometrics + TCCM
Prodan et al. (2024)	Scopus + WoS	747 of 1,024	PRISMA, VOSviewer
Paulovici (2026)	Scopus	892	VOSviewer
This study	**5 databases + central banks**	**4,710**	**+ text as data**

- Gap: one or two databases, mostly English journal articles; central banks absent; who writes is not checked.
- Maps say *what* is studied, not *how* (tone, stance) or *why*; no measure is validated.
- Here: economists only (checked text by text), all languages, tone and stance over time, every coder validated.

Found in our own corpus: 29 titles announce a literature, systematic or bibliometric review of CBDCs (2019–2026).

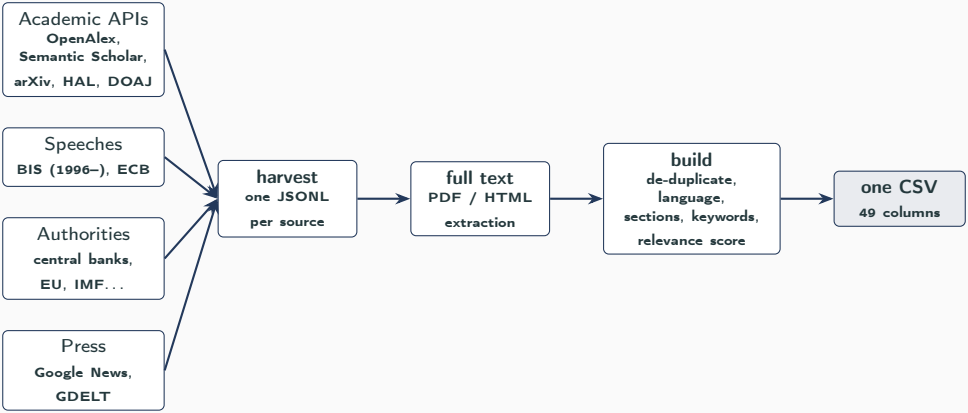
Approach: one tool per question

Question	Tool	Why this one	Explained
Who?	co-citation, co-authorship (association strength, Louvain); laws; log-odds with Dirichlet prior	standard science maps; log-odds shrinks rare words	slide 39, slide 42
What?	thematic map; BERTopic (SBERT, UMAP, HDBSCAN); labels by c-TF-IDF	groups texts by meaning, across languages	slide 43, slide 44
How?	tone: FinBERT on CBDC sentences ($\kappa = 0.52$); stance: reading + local LLM ($\kappa = 0.45$)	best of the coders tested against reading	slide 48, slide 49
Why?	validated codebook ($\kappa = 0.46$) + covariate logit	beats SBERT, NLI, LLMs ($\kappa \leq 0.25$)	slide 47, slide 46
When?	Kleinberg bursts; event studies with placebo dates; difference-in-differences	placebo fixes false positives (13%)	slide 50, slide 51, slide 53
All	7 robustness specifications; every coder checked on hand-coded texts		slide 54

Every result slide links to the slide that explains its method (appendix).

Data

Pipeline



- 115 search phrases, 12 languages: *CBDC, digital euro, e-CNY, eNaira, Drex, digital fiat currency, Fedcoin, euro numérique, digitaler Euro, digital ruble...*
- Recall first: collect everything, then filter in four steps: homonyms, centrality, labels, economists.

Step 1: is it really *Central Bank Digital Currency*?

```
TERMS = {
  "CBDC": r"\bCBDCs?\b|\bwCBDCs?\b|\brCBDCs?\b",
  "central bank digital currency":
      r"central[\s-]+banks?(?:['']s?)?[\s-]+"
      r"digital[\s-]+currenc(?:y|ies)",
  "digital euro": r"digital(?:e[rn]?)?\s+euro\b"
      r"|euro\s+numérique|euro\s+digitale?\b",
  "DCEP (e-CNY)": r"\bDC/?EP\b",
  ...
}
```

- "CBDC" is also a *skin disease*; "DCEP" a chemical.
- Ambiguous terms $t \in \mathcal{A}$ count only with monetary context:

$$\text{count}(t) = n_t \cdot 1\big[t \notin \mathcal{A} \vee \text{context}\big]$$

- context = *currency, payment, central bank, monnaie, Geld...*
- *digital dollar*, *Sand Dollar*: reported, never scored.

Step 2: how central is CBDC to the text?

Each available field f gets a centrality $c_f \in [0, 1]$:

$$c_{\text{title}} = 1[\text{CBDC term in title}] \qquad c_{\text{kw}} = 0.4 \cdot 1[\text{CBDC in keywords}]$$

$$c_{\text{abs}} = \frac{\#\{\text{sentences mentioning CBDC}\}}{\#\{\text{sentences}\}} \qquad c_{\text{body}} = \tfrac{1}{2} \min\left(1, \frac{\rho}{3}\right) + \tfrac{1}{2}\, \sigma$$

ρ = mentions per 1,000 words; σ = share of the ten slices of the text that mention CBDC.

$$\boxed{\text{score} = 100\left(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\right)} \qquad F = \text{fields available}$$

One field clearly about CBDC suffices; a passing mention stays low; missing fields are not penalised.

Step 3: labels, and how good is the filter?

Label (all texts kept)	Texts	Theme	95% CI
Main theme (≥ 50)	15,464	29/30	0.83–0.99
Secondary theme (25–49)	2,245	11/29	0.23–0.56
Mentioned only (< 25)	489	1/30	0.01–0.17
Press, text not available	13,446	2/19	0.03–0.31
Ambiguous terms only	1,709	3/10	0.11–0.60
Off-topic (homonyms, no match)	14,803	0/20	0.00–0.16
Dated before 2014 (checked: false)	32	–	–

Theme = share read as having CBDC as a theme, in a stratified sample of 138 texts read blind to the label; Wilson intervals.

Kept: main + secondary theme (17,709 texts).

Strata h weighted by size N_h; T = theme labels:

$$\widehat{P} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h \in T} N_h}, \qquad \widehat{R} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h} N_h \hat{p}_h}$$

- Precision 0.89, recall 0.89.
- Secondary theme is noisy $\Rightarrow$ robustness on main theme only.

Step 4: is it written by an economist?

Texts with CBDC as a theme	Texts	
Research, classified in economics or finance (OpenAlex)	3,783	kept
Research, other field but an author at a central bank, IMF or BIS	148	kept
Research without OpenAlex record, economics venue	63	kept
Research in other fields	3,469	dropped
Central bank speeches	511	kept
Documents of central banks and international institutions	205	kept
Documents of governments and parliaments	34	dropped
Press articles (no author)	9,496	dropped

Field = OpenAlex primary topic ("Economics, Econometrics and Finance"); institutions = authors' affiliations in OpenAlex. Research dropped: Computer Science 1,769, Business, Management and Accounting 663, no OpenAlex record 473, Social Sciences 281.

- 4,710 texts by economists: research economists, and central banks and international institutions (staff economists, institutional voice).
- Press dropped: no author is known, and journalists are not economists.

▷ method explained: slide 38

Sources and completeness (economists' corpus)

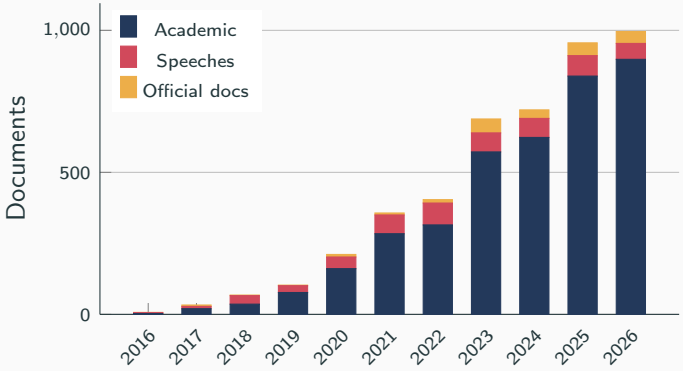
Database	Documents
OpenAlex	4,043
Semantic Scholar	2,164
BIS central bankers' speeches	465
DOAJ	251
Central banks & authorities (crawl)	205
ECB speeches	108
HAL	31
arXiv	24

Merged duplicates count in each database.

Type of text	Documents
Academic	3,994
Speeches	511
Official docs	205
Total	4,710

Field available	Share
Exact day of publication	71%
Issuer / outlet identified	97%
Abstract	68%
Full text	63%
Keywords	85%
Reference list	32%

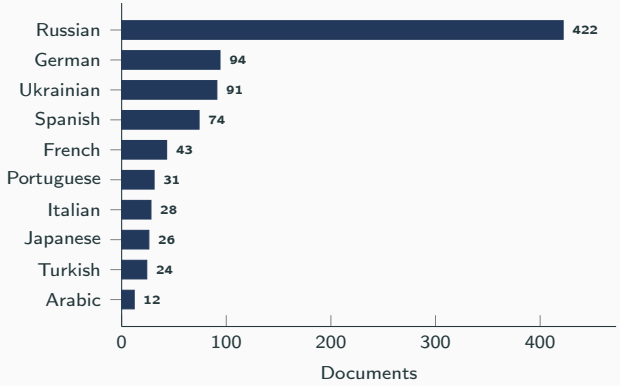
A literature that took off after 2020



- From 103 in 2019 to 956 in 2025.
- Growth 60% a year.
- Turning point 2020: Libra (2019), COVID, the ECB report (Oct. 2020).

0 documents before 2016; 2026 runs to October.

Languages



- English dominates: 3,721 texts (79%); the chart shows the others.
- Second language: Russian (422 texts).
- Everything kept in its original language: the topic model and the codebook are multilingual.

Who writes

Research economists: outlets, most cited, laws

Main outlets

SSRN Electronic Journal	461
RePEc: Research Papers in Economics	164
Zenodo (CERN repository)	126
CyberLeninka (Russian repository)	70
Springer eBooks	68
Routledge eBooks	56
Ekonomika i Upravlenie	47

Bibliometric laws. Growth 60%/year (2017–2025). Lotka: $f(n) \propto n^{-3.3}$, 83% of authors publish once. Bradford: 21 core sources hold 1/3 of the papers (zones 21 : 310 : 1,148).

Most cited papers (global citations)

Bordo (2017)	Central Bank Digital Currency and the Future of Monetary...	517
Auer (2020)	Rise of the Central Bank Digital Currencies: Drivers, App...	454
Agur (2019)	Designing Central Bank Digital Currencies	433
Andolfatto (2020)	Assessing the Impact of Central Bank Digital Currency on...	404
Barrdear (2016)	The Macroeconomics of Central Bank Issued Digital Currenc...	341

Working papers (SSRN, RePEc) weigh as much as journals; a young, open field: most authors write once.

▷ method explained: slide 41

Who speaks: central banks, institutions, research centres

Speeches, by central bank

European Central Bank	118
Deutsche Bundesbank	63
Bank of France	49
Hong Kong Monetary Authority	23
Federal Reserve Board	21
Bank of England	21
Bank of Italy	19
Bank of Japan	14

Documents, by institution

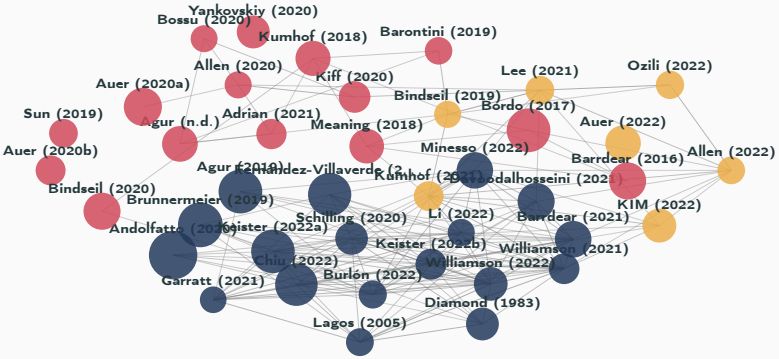
European Central Bank	98
Bank of Japan	27
Sveriges Riksbank	22
Deutsche Bundesbank	21
Swiss National Bank	9
Central Bank of The Bahamas	7
Bank of Jamaica	6
Bank of Canada	4

Research, by author institution

Financial University	63
European Central Bank	61
International Monetary Fund	39
Bank of Canada	38
The Russian Presidential Acad...	33
Bank for International Settle...	32
Moscow State Institute of Int...	29
Lomonosov Moscow State Univer...	29

European Central Bank is the most vocal central bank: the digital euro makes it the hub of the debate.

Co-citation: the intellectual base



Most cited within the corpus

Andolfatto (2020)	264
Brunnermeier (2019)	218
Bordo (2017)	204
Agur (2019)	204
Keister (2022)	203
Fernández-Villaverde (2020)	199
Chiu (2022)	191
Auer (2020)	144

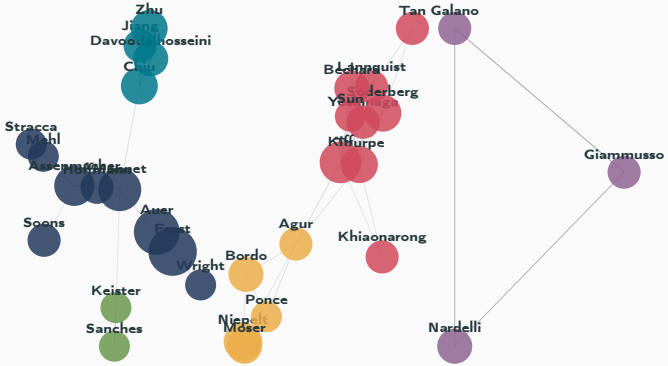
Clusters (largest members):

- Andolfatto (2020), Brunnermeier (2019), Agur (2019), Keister (2022a)
- Bordo (2017), Auer (2020a), Barrdear (2016), Bindseil (2020)
- Auer (2022), KIM (2022), Kumhof (2021), Lee (2021)

1,493 papers with reference lists, 13,810 distinct references; 40 most co-cited shown, 3 clusters, $Q = 0.10$. ▷ method

explained: slide 39

Co-authorship: research communities



Most productive authors

Ulrich Bindseil	20
Peterson K Ozili	20
Jon Frost	16
Annelieke Mooij	16
Raphael Auer	14
Cyril Monnet	12
John Kiff	11
Katrin Assenmacher	10

5,716 authors; 48% of papers co-authored (mean team 1.9); 15% international. Top countries: RU 473, US 248, DE 167, CN 166, GB 153.

Papers with a central bank, BIS, IMF or World Bank author: 10% (23% in 2014–2019, 7% in 2023–2026), but 48% of citations.

▷ method explained: slide 39

Who says what: research economists vs central bankers

Words most over-used by each group compared with the other (z-scores):

Research economists (259,105 words)		**Central bankers** (232,446 words)	
legal	23	payments	28
analysis	21	euro	22
development	21	european	18
research	20	europe	18
monetary	20	area	18
impact	19	services	15
model	18	innovation	14
implementation	17	conference	14

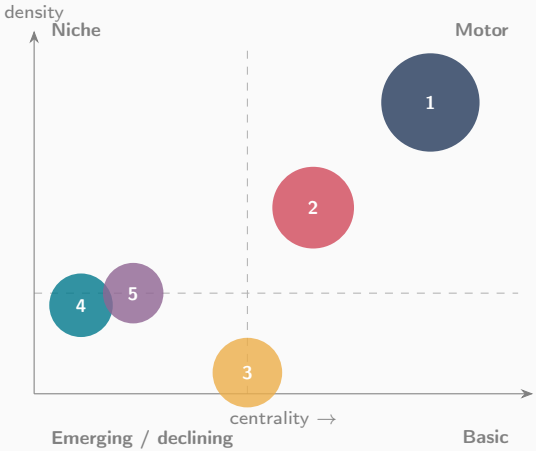
English texts; research: title + abstract; central bankers: speeches; topic words (*CBDC, digital, currency...*) removed.

Reading: research economists: *legal, analysis, development*; central bankers: *payments, euro, european*.

▷ method explained: slide 42

What is written

Thematic map: which themes drive the field?



Keyword clusters placed by centrality (links to other themes) and density (internal cohesion).

	Theme (top keywords)	**Quadrant**
1	digital transformation in financial services, monetary policy, moneta. . .	motor
2	blockchain technology applications and security, digital transformati. . .	motor
3	fintech, crowdfunding, digital finance, blockchain, fintech	basic
4	global financial crisis and policies, international payments, stablec. . .	emerging or declining
5	financial inclusion, digital payments, india	niche

Motor themes: digital transformation in financial services; blockchain technology applications and security.

▷ method explained: slide 40

How the themes evolve

Keyword (% of the period's papers) *papers in period*	**2014–2019** *136*	**2020–2022** *755*	**2023–2026** *2,919*	**Change**
financial inclusion	5.9	8.2	15.8	+7.6
digital euro	2.9	7.8	11.4	+3.6
global financial crisis and policies	2.2	8.7	11.8	+3.0
financial stability	24.3	12.6	15.6	+3.0
fintech, crowdfunding, digital finance	48.5	48.7	49.8	+1.0
digital ruble	0.0	13.9	14.8	+0.9
european monetary and fiscal policies	16.9	10.3	10.3	−0.1
digital transformation in financial ser. . .	72.1	74.7	74.2	−0.5
digital transformation in law	10.3	17.9	17.3	−0.6
banking stability, regulation, efficien. . .	22.1	16.4	15.5	−0.9
monetary policy and economic impact	51.5	23.0	20.7	−2.4
digital payments	8.8	12.6	10.2	−2.4

Sorted by change between the last two periods. Darker = more frequent.

New in each period (c-TF-IDF, periods as classes): **2014–2019**: *e-krona, riksbank, issue, legal, innovation, services*; **2020–2022**: *euro, ruble, china, legal, interest, national*; **2023–2026**: *euro, legal, ruble, regulatory, india, data*.

Rising fastest: financial inclusion; falling most: digital payments.

▷ method explained: slide 45

Topics found by meaning (SBERT + BERTopic)

	Topic (top c-TF-IDF terms)	**Texts**	**% of the period's texts**			**Adjusted, points**	
			2014–2019	**2020–2022**	**2023–2026**	**change**	**CB**
1	cryptocurrencies, implementation, private, tr...	470	18.5	13.5	8.5	**-13.2**	**-8.6**
2	design, retail, legal, private	439	10.4	11.2	8.6	-4.0	**-9.4**
3	uncertainty, rates, design, rate	349	14.2	8.7	6.7	**-9.6**	**-5.9**
4	dollar, trade, debt, reserve	326	5.7	5.8	7.5	+0.8	**-5.0**
5	deposits, interest, deposit, rate	319	2.8	5.0	7.3	+3.6	**-7.7**
6	governor, innovation, fintech, kong	306	17.5	11.2	4.7	-1.2	**+35.7**
7	russian, russia, ruble, federation	238	0.0	5.1	4.9	**+4.7**	**-5.2**
8	china, rmb, e-cny, yuan	213	1.4	5.3	4.6	+2.7	**-5.2**
9	cashless, evolution, riksbank, forms	204	6.6	4.9	4.1	**-3.8**	**-1.8**
10	fintech, technologies, services, innovation	199	7.1	2.9	4.4	**-3.0**	-0.5
11	euro, european, ecb, area	186	0.5	3.6	4.2	**+3.7**	**+5.6**
12	cryptocurrencies, crypto, bitcoin, cryptocurr...	186	9.0	4.0	3.6	**-5.8**	-1.1

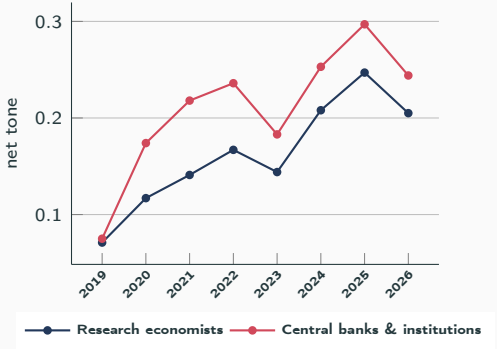
4,710 texts; 24 topics (50% outliers reassigned). Adjusted: logit on period, actor, English; change = 2023–26 vs 2014–19; CB = central banks vs research; **bold**: 95% CI excludes 0.

Topics tied to one jurisdiction: 6–34% of texts in every period, but the mix shifts: fall; *russian*, *euro*, *india* rise.

▷ method explained: slide 44

How: tone and stance

Tone: how positive is what economists say about CBDCs?



	Texts	Sent.	%+	%−	Net tone
Research economists	2,306	7,519	21	5	0.19 ± 0.01
Central banks & institutions	630	7,576	18	3	0.22 ± 0.01

FinBERT on the 15,095 English sentences that name a CBDC. Sentence tone $= P(+) - P(-)$, averaged per text, then per group ($\pm$ 95% CI). 90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

Reading: tone rises steadily in both groups; research economists from 0.07 in 2019 to 0.25 in 2025; central banks stay slightly more positive.

▷ method explained: slide 48

Do policy events move the tone?

Date	Event	Before	After	Δtone	p
Jun 2019	Facebook announces Libra	23	42	-0.041	0.33
Oct 2020	ECB digital euro report	46	113	+0.003	0.89
Oct 2020	Bahamas launches Sand Dollar	57	111	-0.020	0.55
Jul 2021	ECB investigation phase	110	102	+0.020	0.56
Oct 2021	Nigeria launches eNaira	115	103	+0.050	0.24
Jan 2022	Fed CBDC discussion paper	102	122	+0.022	0.51
Mar 2022	US order on digital assets	108	129	+0.004	0.87
Nov 2022	India e-rupee pilot	130	128	+0.011	0.71
Feb 2023	UK digital pound consultation	132	148	-0.008	0.77
Jun 2023	EU digital euro bill	145	211	-0.043	0.30
Oct 2023	ECB preparation phase	169	217	-0.025	0.48
Jan 2025	US order banning a US CBDC	193	230	+0.018	0.60

For each event: mean tone of the texts in the 180 days *after* minus the 180 days *before*. Compared with 500 random dates: p = share of random dates with a larger $|\Delta|$ (95th percentile: 0.084).

Result: 0 of 12 events move the tone (smallest $p = 0.24$).

Tone drifts, it does not jump: no single event explains the rise.

▷ method explained: slide 52

Stance: for or against CBDCs?

Level, read by hand, % [95% CI]

	n	Favourable	Critical
Research economists	50	8 **[3–19]**	2 **[0–10]**
Central banks & institutions	50	52 **[39–65]**	6 **[2–16]**

By reading: 52% of central bank texts are favourable; research economists are mostly neutral (90%): they analyse, central banks advocate.

Change (local LLM, 575 texts sampled per actor × period), % favourable

	2014–2019	2020–2022	2023–2026	Change, pts
Research economists	26	38	40	**+14** **[+1, +27]**
Central banks & institutions	25	54	65	**+40** **[+26, +53]**

The LLM ($\kappa = 0.45$ on 100 economists' texts, best of four coders, slide 56) calls 46% of the hand-coded texts favourable against 30% by reading: its levels are inflated, so only its *changes* are used. **Bold**: CI excludes 0.

Reading: both groups become more favourable; central banks much more, as their own projects advance.

▷ method explained: slide 49

Why: stated motivations

Coding stated motivations

```
MOTIVES = {
  "Financial inclusion":
      r"financial\s+inclusion|unbanked|underbanked"
      r"|access\s+to\s+(?:financial\s+services|finance)",
  "Monetary sovereignty":
      r"monetary\s+sovereignty|global\s+stablecoin"
      r"|\blibra\b|big\s*tech|dollari[sz]ation",
  "Decline of cash":
      r"declin\w*\s+(?:in|of)\s+(?:the\s+)?cash"
      r"|cashless|use\s+of\s+cash",
  ...
}
```

- Motivations are stated in the introduction (else abstract, else opening).
- 10 motivations; codebook built by reading introductions; several per text allowed.
- 4,110 texts coded; 76% state at least one.
- Validated on the 50 held-out texts by economists (of 74; reading-based coding, AI-assisted): precision 0.70, recall 0.41, $\kappa = 0.46$.

▷ method explained: slide 47

Why: what economists say CBDCs are for

Motivation	% of texts 2014–2019	% of texts 2020–2022	% of texts 2023–2026	Adjusted effect, points [95% CI] Change 2023–26 vs 2014–19	Adjusted effect, points [95% CI] Central banks vs research
Financial stability	30.8	26.3	29.3	-1.6 [-8, +5]	**-7.7 [-11, -4]**
Monetary policy	36.8	27.4	24.4	**-15.1 [-22, -8]**	**-13.6 [-16, -11]**
Technology (DLT, programmability)	22.5	21.6	25.8	+1.6 [-5, +8]	**-9.1 [-12, -6]**
Cross-border payments	13.2	19.0	22.5	**+8.7 [+3, +13]**	**-7.8 [-11, -5]**
Payment efficiency	10.4	12.1	17.4	**+6.6 [+1, +11]**	**-3.1 [-6, -0]**
Monetary sovereignty	15.9	16.8	13.6	-2.0 [-8, +3]	-2.4 [-5, +0]
Financial inclusion	3.8	8.9	16.8	**+12.3 [+8, +15]**	**-9.5 [-12, -7]**
Geopolitics and sanctions	2.7	7.2	15.0	**+12.4 [+9, +15]**	+0.3 [-2, +3]
Privacy and surveillance	7.1	10.5	13.6	**+6.4 [+2, +10]**	**-2.8 [-5, -0]**
Decline of cash	14.8	9.0	7.4	**-6.5 [-12, -2]**	+1.2 [-1, +4]

Logit of each motivation on period, actor and English; average marginal effects, robust SEs. **Bold**: CI excludes 0.

- Rising: geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts). Falling: monetary policy (−15 pts), decline of cash (−7 pts).
- From a monetary-policy instrument to a tool for inclusion, payments and geopolitics. Recall ≈ 0.41: shares are lower bounds; compare motivations and periods rather than read levels.

▷ method explained: slide 46

Is the codebook the best coder?

Coder	P	R	F_1	κ	κ, all
Codebook (regex)	0.70	0.41	0.52	0.46	0.50
Codebook OR classifier	0.42	0.60	0.50	0.39	0.42
SBERT, classifier (100 labels)	0.36	0.37	0.37	0.25	0.28
Local LLM, Qwen2.5 7B	0.30	0.61	0.40	0.25	0.28
Local LLM, Qwen2.5 3B	0.31	0.23	0.26	0.15	0.16
SBERT, definitions (zero-shot)	0.21	0.71	0.32	0.12	0.15
NLI zero-shot (mDeBERTa)	0.22	0.39	0.28	0.11	0.14

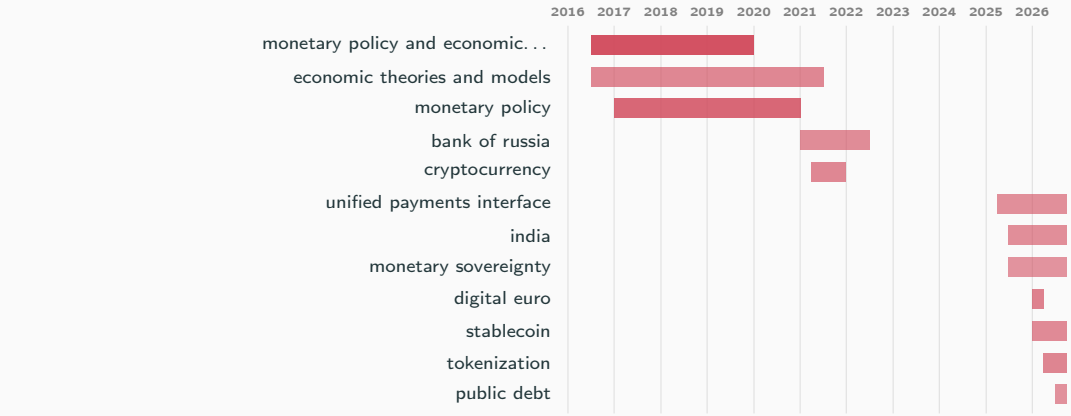
The 50 held-out texts written by economists $\times$ 10 motivations; last column: κ on all 74 held-out texts. Thresholds and classifiers set on 100 development texts.

- The codebook wins: $\kappa = 0.46$; best alternative 0.25.
- Models confuse "mentions privacy" with "studies CBDCs because of privacy".
- Trends hold: same direction of change for 9 of 10 motivations with the high-recall coding, 9 of 10 on main-theme texts only.

▷ method explained: slide 47

When: the environment

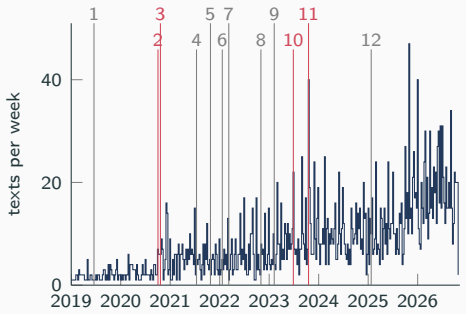
Which themes burst, and when



Reading: early bursts on monetary policy and economic impact, economic theories and models, monetary policy; the latest on unified payments interface, india, monetary sovereignty, digital euro. ▷

method explained: slide 50

Attention around policy events



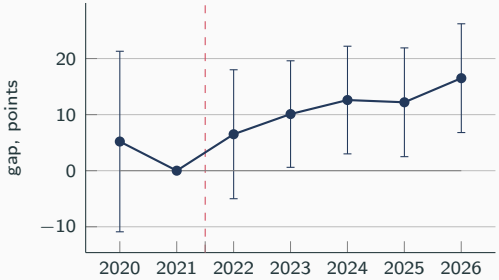
	Date	Event	Ratio	z	p
1	Jun 2019	Facebook announces Libra	0.3	-1.3	0.83
2	Oct 2020	ECB digital euro report	2.9	+4.0	0.01
3	Oct 2020	Bahamas launches Sand Dollar	3.0	+4.7	0.01
4	Jul 2021	ECB investigation phase	0.5	-1.9	0.93
5	Oct 2021	Nigeria launches eNaira	0.8	-0.7	0.69
6	Jan 2022	Fed CBDC discussion paper	1.4	+1.2	0.24
7	Mar 2022	US order on digital assets	0.6	-1.3	0.84
8	Nov 2022	India e-rupee pilot	1.0	-0.2	0.55
9	Feb 2023	UK digital pound consultation	0.7	-1.0	0.76
10	Jun 2023	EU digital euro bill	1.7	+2.9	0.05
11	Oct 2023	ECB preparation phase	3.3	+10.0	0.00
12	Jan 2025	US order banning a US CBDC	1.2	+0.9	0.29

Texts in the 15 days after the event vs. the rate of the 3 months before. p = share of 500 random dates with a larger z (the naive test flags 13% of them).

Reading: 4 of 12 events beat 95% of placebo dates; strongest: ECB preparation phase, Bahamas launches Sand Dollar.

▷ method explained: slide 51

The 2022 sanctions: did geopolitics enter the discussion?



Gap in the share stating *geopolitics and sanctions*: texts about Russia (or in Russian) minus the others, relative to 2021; 95% CI.

756 texts about Russia among 3,772 (2020–2026). Pooled effect after 2021, every motivation:

Motivation	DiD, points
Geopolitics and sanctions	**+11.6** [+5, +18]
Cross-border payments	+8.1 [-2, +18]
Payment efficiency	+7.1 [-1, +15]
Decline of cash	+2.8 [-2, +8]
Privacy and surveillance	-4.0 [-11, +3]
Financial inclusion	**-6.7** [-13, -0]
Monetary sovereignty	-8.7 [-18, +1]
Technology (DLT, programmability)	-10.0 [-21, +1]
Monetary policy	**-12.0** [-24, -0]
Financial stability	**-12.6** [-25, -1]

Reading: geopolitics rises by +11.6 points [+5, +18] more in texts about Russia, and the gap widens each year. But 3 other motivations shift too: consistent with a sanctions effect, not proof.

▷ method explained: slide 53

The discussion and its environment

Putting it together, period by period

	2014–2019	2020–2022	2023–2026
Policy events (bold: attention reacts)	**Facebook announces Libra**	ECB digital euro report; Bahamas launches Sand Dollar; **ECB investigation phase**; **Nigeria launches eNaira**; **Fed CBDC discussion paper**; **US order on digital assets**; India e-rupee pilot	**UK digital pound consultation**; EU digital euro bill; ECB preparation phase; **US order banning a US CBDC**
Texts per year	**53**	**324**	**840**
Bursting keywords	**monetary policy and economic impact, economic theories and models, monetary policy**	**bank of russia, cryptocurrency**	**unified payments interface, india, monetary sovereignty**
Most frequent motivations	**monetary policy, financial stability, technology**	**monetary policy, financial stability, technology**	**financial stability, technology, monetary policy**
Net tone (research / central banks)	**0.06 / 0.10**	**0.15 / 0.22**	**0.21 / 0.24**
Favourable, LLM (research / central banks)	**26% / 25%**	**38% / 54%**	**40% / 65%**

- 2014–19: a monetary-policy question, alongside Libra and the decline of cash; few texts, cautious tone.
- 2020–22: central bank reports and pilots; the discussion grows and turns favourable.
- 2023–26: national projects, sanctions, cross-border payments; the most favourable period.

What we can and cannot say about causes

Link	Test	Result	Verdict
Events → attention	event study vs. placebo dates	4 of 12 events	**timing evidence**
Events → tone	180-day windows vs. placebo dates	0 of 12 events	**no effect found**
Sanctions → geopolitics motive	difference-in-differences	+11.6 points [+5, +18]	**consistent with**
Events → stance	–	sample too small around each date	not testable
Motivations ↔ stance	–	measured in the same texts	association only

- The paper stays descriptive: it measures changes and shows what moves with them.
- Explaining *why* economists changed their stance would need other evidence (interviews, authors' careers): hypotheses for further work.

Robustness

Robustness

Specification	Texts	Why: change, points inclusion	Why: change, points monetary policy	How: tone CB − research	How: stance fav. CB − research	When events
Baseline: economists	4,710	+12.3 [+8, +15]	−15.1 [-22, -8]	+0.03	+16	4/12
Main theme only	3,513	+14.9 [+10, +18]	−16.0 [-25, -7]	+0.01	+13	4/12
English only	3,721	+13.7 [+9, +16]	−16.0 [-23, -7]	+0.03	+16	4/12
Journal articles only (research)	3,171	+12.3 [+6, +15]	−16.8 [-25, -8]	+0.01	+11	4/12
Without 2026 (partial year)	3,547	+11.8 [+7, +15]	−15.3 [-23, -8]	+0.03	+15	4/12
High-recall motivation coding	4,710	+15.6 [+10, +19]	−8.8 [-16, -2]	+0.03	+16	4/12
All disciplines (not only economists)	8,213	+13.2 [+10, +15]	−15.0 [-21, -9]	+0.02	+15	4/12

Changes: 2023–26 vs 2014–19, adjusted (logit, 95% CI). Tone: mean net tone gap. Stance: raw LLM labels, points. Events: $p < 0.05$ against placebo dates.

Signs and orders of magnitude hold in every specification, including when research from every discipline is kept.

▷ method explained: slide 54

Conclusion

Takeaways and limits

How economists' discussion of CBDCs evolved

- Who: 4,710 texts; central bank economists write 10% of the papers but get 48% of citations.
- What: from monetary policy and design to inclusion and national projects (digital ruble, rupee, e-CNY).
- How: tone and stance become more favourable; research economists stay mostly neutral, central banks advocate.
- Why: rising geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts); falling monetary policy (−15 pts), decline of cash (−7 pts).
- When: 4 of 12 events move attention, 0 of 12 move tone; geopolitics rises after the 2022 sanctions.

Limits

- Economists identified by field and affiliation, not by training.
- Motivations: recall ≈ 0.5 (lower bounds). Tone: English only. Stance: LLM for changes only.
- One coder (AI-assisted) for the validation samples.
- Descriptive: causes stay hypotheses.

Methods explained

Methods explained: how to read this appendix

One slide per tool, each in four parts:

- **Idea**: what the tool does, in words.
- **How**: the computation, step by step, with its formula.
- **Why this tool**: what it solves, and what it was compared with.
- **Used on**: links to the result slides.

Tool	Slide	Tool	Slide
Who is an economist	slide 38	Covariate-adjusted logit	slide 46
Networks: association strength, Louvain	slide 39	Codebook and validation	slide 47
Thematic map (Callon)	slide 40	Tone (FinBERT)	slide 48
Bibliometric laws	slide 41	Stance (reading + LLM)	slide 49
Log-odds with Dirichlet prior	slide 42	Bursts (Kleinberg)	slide 50
Sentence embeddings (SBERT)	slide 43	Event study and placebo	slide 51
Topic model (BERTopic)	slide 44	Tone around events	slide 52
Class-based TF-IDF	slide 45	Difference-in-differences	slide 53
		Robustness	slide 54

Method: who counts as an economist

Idea

The course asks how *economists* discuss CBDCs, so each text must be traced to an economist or an economic institution.

How

- **Research**: kept if OpenAlex classifies it in *Economics, Econometrics and Finance* (primary topic, from title, abstract, venue, citations), or an author works at a central bank, the IMF or the BIS; without OpenAlex record: an economics venue.
- **Institutions**: central bank speeches; documents of central banks and international financial institutions.
- **Dropped**: other fields (computer science, law…), governments and parliaments, the press.

Why this tool

Authors' training is not recorded anywhere at scale; the field of the paper and the employer of the author are, and they are what defines a discipline in bibliometrics.

Speeches are kept as *institutional* voices: written with the bank's economics staff, even when the speaker is not an economist.

Robustness: results with every discipline kept are on slide 35.

Used on

slide 12, and every result after it.

Method: networks (association strength, Louvain)

Idea

Two items (references, authors, keywords) are linked when they appear in the same paper; groups of tightly linked items are schools, teams or themes.

How

1. Count c_{ij} = papers where i and j appear together; c_i = papers with i.
2. Normalise by association strength $s_{ij} = c_{ij}/(c_i c_j)$: the ratio of observed to expected co-occurrence.
3. Cluster with Louvain: move nodes between groups while the modularity $Q = \frac{1}{2m}\sum_{ij}\left[s_{ij} - \frac{k_i k_j}{2m}\right]\delta(g_i, g_j)$ rises.
4. Draw: force-directed layout; size $\propto \sqrt{c_i}$.

Why this tool

Raw counts favour popular items: two classics are co-cited often by chance. Association strength removes this (van Eck & Waltman, 2009); it is VOSviewer's default.

Louvain is fast and needs no number of clusters; Q tells how clear the grouping is.

Co-citation (references cited together) shows the intellectual base; **co-authorship** shows teams.

Used on

slide 18, slide 19, slide 21.

Method: thematic map (Callon)

Idea

Place each cluster of keywords on two axes: how much it is linked to the rest of the field, and how developed it is inside.

How

With the equivalence index $e_{ij} = c_{ij}^2/(c_i c_j)$:

$$\text{centrality} = 10 \sum_{i \in C,\, j \notin C} e_{ij}, \qquad \text{density} = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}$$

Quadrants, split at the medians:

- **motor**: central and dense; **basic**: central, loose;
- **niche**: dense, isolated; **emerging or declining**: neither.

Why this tool

A keyword list says what is frequent; the map says what *structures* the field. It is the standard tool of science mapping (Callon et al. 1991; Cobo et al. 2011; bibliometrix).

Limit: keywords are partly machine-assigned, so the topic model (meaning-based) is used alongside.

Used on

slide 21.

Method: bibliometric laws

Idea

Three regularities describe the shape of any scientific field: how fast it grows, how output is spread among authors, and among journals.

How

- **Growth**: compound annual rate $g = (N_{y_1}/N_{y_0})^{1/(y_1 - y_0)} - 1$.
- **Lotka**: share of authors with n papers $f(n) \propto n^{-\beta}$; β fitted on the log-log counts ($\beta \approx 2$ in mature fields).
- **Bradford**: sort sources by output, cut in three zones holding $1/3$ of papers each; the first zone is the core.

Why this tool

They situate the field: a high growth rate and a large β (most authors write once) mean a young, open field around a small core of outlets.

They are the descriptive baseline of every bibliometric review, so our numbers can be compared with the earlier reviews.

Used on

slide 14, slide 16.

Method: distinctive words (log-odds with Dirichlet prior)

Idea

Which words does one group use *more* than the other, once chance is accounted for?

How

For word w, group i vs. the other group j, with counts y, totals n, and a prior $\alpha_w = \alpha_0 \times$ (corpus frequency of w):

$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}$$

$$z_w = \delta_w \Big/ \sqrt{\frac{1}{y_{iw}+\alpha_w} + \frac{1}{y_{jw}+\alpha_w}}$$

Words are ranked by z_w.

Why this tool

Plain frequencies list words everyone uses; TF-IDF or raw log-ratios are dominated by rare words (a word used twice by one group gets an infinite ratio). The prior shrinks rare words towards zero and the z-score accounts for uncertainty (Monroe, Colaresi & Quinn, 2008).

Used on

slide 20.

Method: sentence embeddings (SBERT)

Idea

Turn each text into a vector of 384 numbers such that texts with similar *meaning* get close vectors, whatever their words or language.

How

- A transformer network reads title + abstract (or opening) and averages its word representations into one vector $\mathbf{e}_t$.
- Model: `paraphrase-multilingual-MiniLM-L12-v2`, trained so that translations and paraphrases land together (50+ languages).
- Similarity: cosine $\cos(\mathbf{e}_s, \mathbf{e}_t) = \frac{\mathbf{e}_s \cdot \mathbf{e}_t}{\|\mathbf{e}_s\| \|\mathbf{e}_t\|}$.

Why this tool

Keywords and word counts treat "digital ruble" and its Russian original (*tsifrovoy rubl'*) as unrelated; embeddings do not. With 79% English and many Russian texts, this is what makes a single topic model possible.

Tested also as a motivation coder (similarity to prototypes): it lost to the codebook (slide 29).

Used on

slide 23.

Method: topic model (BERTopic)

Idea

Find groups of texts about the same subject, without fixing the subjects in advance.

How

1. **Embed** each text (SBERT).
2. **Reduce** 384 dimensions to 5 with UMAP, which keeps neighbours together.
3. **Cluster** with HDBSCAN: dense regions become topics (at least 40 texts; leaf selection, which favours fine topics over one large blob); texts in no dense region are outliers, then assigned to the nearest topic.
4. **Label** each topic by its c-TF-IDF terms (slide 45).

Why this tool

LDA and STM count words, one language at a time, and need the number of topics. BERTopic works on meaning, across languages, and finds the number itself (Grootendorst, 2022).

The covariate part of STM is kept: topic shares are adjusted for period, actor and language (slide 46).

Used on

slide 23.

Method: class-based TF-IDF (c-TF-IDF)

Idea

Which words characterise a class of texts (a topic, or a period) compared with all the others?

How

Merge the texts of each class c into one document; for term t:

$$W_{t,c} = \frac{\mathrm{tf}_{t,c}}{|c|} \log\left(1 + \frac{A}{f_t}\right)$$

$\mathrm{tf}_{t,c}$ = count of t in class c, $|c|$ = words in c, A = average words per class, f_t = count of t in all classes. Words in more than 15% of texts and generic words are removed.

Why this tool

Ordinary TF-IDF scores words within one document; here the unit is a whole class. A frequent word in one class but rare elsewhere scores high: it is what is *new* or *specific*.

Periods as classes answer "when": which vocabulary appears in each period.

Used on

slide 23 (topic labels), slide 22 (periods).

Method: covariate-adjusted logit

Idea

A raw change between periods can come from a change in *who* writes (more central banks, more Russian texts). Compare like with like.

How

For each outcome $y_t \in \{0, 1\}$ (a motivation, a topic):

$$P(y_t = 1) = \Lambda\big(a + b_{\text{period}} + c_{\text{actor}} + e \cdot \text{English}\big)$$

- **Average marginal effect**: predict every text in 2023–26, then in 2014–19, average the difference (points).
- 95% CI: robust (HC1) standard errors; coefficients drawn 400 times from their estimated distribution.

Why this tool

It is the "prevalence" model of a structural topic model (Roberts et al., 2014), applied to any coded outcome. Marginal effects are read in percentage points, unlike log-odds coefficients.

Used on

slide 28, slide 23, slide 35.

Method: codebook and validation

Idea

Write down, for each of 10 motivations, the phrases authors use to state it; flag a text when its introduction contains one.

How

1. Read 100 introductions (**development set**); write regular expressions in 7 languages.
2. Code 74 *other* texts by reading (**test set**), never used for tuning; scores on the 50 written by economists.
3. Compare: precision $= \frac{TP}{TP+FP}$, recall $= \frac{TP}{TP+FN}$, Cohen's $\kappa = \frac{p_o - p_e}{1 - p_e}$ (agreement beyond chance).

Why this tool

Motivations need reading; a codebook makes the reading explicit, reproducible and checkable.

Five machine coders were tested on the same texts (SBERT $\times$ 2, NLI mDeBERTa, Qwen2.5 3B and 7B): see slide 29.

High precision, recall ≈ 0.5: shares are lower bounds; changes are robust (slide 29).

Used on

slide 27, slide 28, slide 29.

Method: tone (FinBERT)

Idea

Measure how positive or negative the sentences *about CBDCs* are.

How

1. Keep the English sentences that name a CBDC (same term list as the filter).
2. FinBERT (a BERT model trained on financial news) gives $P(+), P(0), P(-)$ for each sentence.
3. Sentence tone $= P(+) - P(-) \in [-1, 1]$; averaged per text, then per group and year.
4. Validation: 90 sentences from economists' texts read blind; $\kappa = 0.52$ (all 0.53).

Why this tool

Dictionary methods (Loughran–McDonald) miss context ("not risky"); general sentiment models misread finance ("liability" is not negative). FinBERT is the standard for economic text (Araci, 2019).

Scoring only CBDC sentences isolates the tone *about CBDCs*, not about the economy.

Limit: it sees tone in neutral sentences, so compare groups and years, not levels.

Used on

slide 24, slide 25.

Method: stance (reading + local language model)

Idea

Tone is about words; stance is about the *author's position*: favourable, neutral or critical towards issuing a CBDC.

How

- **Levels**: texts read by hand per actor; share with a Wilson interval $\frac{\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}}{1+z^2/n}$.
- **Changes**: Qwen2.5 7B, run locally, labels a random sample of up to 100 texts per actor × period; change $= \hat{p}_{\text{late}} - \hat{p}_{\text{early}}$, CI $\pm 1.96\sqrt{\sum \hat{p}(1-\hat{p})/n}$.

Why this tool

Stance needs understanding of the argument: only a language model can do it at scale. Four coders were tested (slide 56); the LLM is best ($\kappa = 0.45$) but over-calls "favourable".

A constant bias cancels in a difference: so the LLM gives *changes*, reading gives *levels*.

Used on

slide 26, slide 33.

Method: bursts (Kleinberg)

Idea

Let the data say when a keyword suddenly becomes frequent, instead of choosing dates.

How

For keyword w, quarter t: r_t of d_t papers use it.

- Two states: normal rate $p_0 = R/D$ (overall), burst rate $p_1 = 2p_0$.
- Cost of a quarter in state i: $\sigma_i(t) = -\ln\left[\binom{d_t}{r_t} p_i^{r_t}(1-p_i)^{d_t - r_t}\right]$; entering a burst costs $\ln T$.
- The cheapest sequence of states (Viterbi algorithm) gives the bursts; weight = cost saved.

Why this tool

A simple growth rate reacts to noise; the entry cost means only sustained rises count. It is the standard "burst detection" of CiteSpace and science-of-science work (Kleinberg, 2002).

Used on

slide 30, slide 33.

Method: event study with placebo dates

Idea

Does writing on CBDCs jump right after a policy event, more than on an ordinary day?

How

- Baseline rate λ = texts per day over days $[-90, -15]$ before the event.
- Expected over $[0, 14]$: $E = 15\lambda$; ratio $= N/E$; $z = (N - E)/\sqrt{E}$.
- Placebo: the same z on 500 random dates; p = share of random dates with a larger z.

Why this tool

The usual Poisson test assumes a stable rate; attention trends upward and comes in waves, so 13% of random dates look "significant". Comparing with random dates gives the right yardstick (as in finance event studies, MacKinlay 1997).

It is causal only in a narrow sense: timing of attention.

Used on

slide 31, slide 34.

Method: do events move the tone?

Idea

Same logic as the event study, applied to the tone of the texts rather than their number.

How

- Texts dated to the day, with a tone score.
- Δ = mean tone over $[t_0, t_0 + 180)$ minus mean tone over $[t_0 - 180, t_0)$.
- Same Δ on 500 random dates; p = share with $|\Delta|$ at least as large (two-sided).

Why this tool

Tone rises over time anyway; a before/after difference alone would credit every event with that trend. Random dates carry the same trend, so only a jump beyond it counts.

180 days: enough texts on each side (100+).

Used on

slide 25, slide 34.

Method: difference-in-differences

Idea

If the 2022 sanctions pushed economists to discuss geopolitics, texts about Russia should change *more* than the others after 2022.

How

Linear probability model, robust SEs; y = motivation stated, T = text about Russia or in Russian:

$$y_{it} = a_T + b_t + \sum_{k \neq 2021} \gamma_k \, T_i \, 1[t = k] + \varepsilon_{it}$$

γ_k = gap in year k relative to 2021. Pooled: one γ for $t > 2021$.

Why this tool

A simple before/after mixes the sanctions with everything else that happened in 2022; the other texts serve as the control.

Two checks: **pre-trends** ($\gamma_k \approx 0$ before 2022: only 2020 is usable) and **placebo outcomes** (other motivations should not move: some do). Hence "consistent with", not proof.

Used on

slide 32, slide 34.

Method: robustness

Idea

Re-run the headline results on samples that remove one possible source of error at a time.

How

- **Main theme only**: the secondary theme label is noisy.
- **English only**: language mix changes over time.
- **Journal articles only**: working papers and chapters differ.
- **Without the current year**: partial year.
- **High-recall coding**: codebook + SBERT matches.
- **All disciplines**: the economist filter itself.

Why this tool

Each measure has a known weakness; a result that holds whichever way the sample is cut does not depend on it. It answers a referee's "what if" in one table.

Used on

slide 35.

Backup

Backup: validation of the motivation coding

Codebook tuned on 100 texts (development set), then evaluated on the 50 held-out texts written by economists (of 74); unit = first 1,500 characters of the introduction / abstract.

Motivation	Read	Coded	Precision	Recall	κ
Privacy and surveillance	3	2	1.00	0.67	0.79
Geopolitics and sanctions	7	4	1.00	0.57	0.70
Technology (DLT, programmability)	11	11	0.73	0.73	0.65
Financial inclusion	7	3	1.00	0.43	0.56
Monetary policy	7	9	0.56	0.71	0.55
Cross-border payments	1	3	0.33	1.00	0.48
Financial stability	10	7	0.57	0.40	0.37
Monetary sovereignty	6	1	1.00	0.17	0.26
Decline of cash	7	1	1.00	0.14	0.22
Payment efficiency	16	3	0.67	0.12	0.12
All	75	44	0.70	0.41	0.46

Read = texts where the reading found the motivation; Coded = texts flagged by the codebook. Files: `presentation/validation_motivations_dev.csv`, `_test.csv`.

Backup: validation of the stance coders

Each coder against the stance read by hand (favourable / neutral / critical); thresholds and classifiers by 5-fold cross-validation.

Coder	**Accuracy**	κ	κ, **all**
Local LLM, Qwen2.5 7B	0.70	0.45	0.41
FinBERT tone	0.65	0.31	0.20
NLI zero-shot (mDeBERTa)	0.63	0.29	0.18
SBERT classifier	0.61	0.21	0.17

Scores on the 100 texts written by economists; last column: all 177 hand-coded texts (including the press).

Files: `presentation/validation_stance.csv`, `validation_predictions_stance.csv`.

Backup: validation of the tone measure

Sentences sampled per FinBERT label, read blind to the label; table: the 90 from economists' texts (30 per label).

FinBERT ↓ / reading →	**negative**	**neutral**	**positive**	**Precision**
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

File: `presentation/validation_tone.csv`.

Backup: how the relevance score behaves

Example	Score	Label
Paper, CBDC in title	87	main theme
Paper, generic title, CBDC abstract	80	main theme
Paper, one passing mention	11	mentioned only
Speech, one paragraph on it	18	mentioned only
Speech devoted to it	80	main theme
Medical case report	0	off-topic

Backup: press, recursive bisection of time windows

```
stack = monthly_windows("2000-01-01", today)
while stack:
    a, b = stack.pop()
    items = fetch(query, after=a, before=b)
    if len(items) >= CAP and (b - a).days > 1:
        m = a + (b - a) / 2
        stack += [(m, b), (a, m)]
        continue
    save(items)
    checkpoint(query, a, b, len(items))
```

A news search returns at most $C = 100$ items. A window holding more is cut in two until each holds fewer, down to one day: depth ≤ 5 per month, $2L - 1$ requests for L windows. The press was collected, then left out of the economists' corpus.

Backup: academic sources, cursor pagination under a rate limit

```
def iter_phrase(phrase):
    cursor = "*"
    while cursor:
        d = get(f'{API}?filter=title_and_abstract'
                f'.search:"{phrase}"&per-page=200'
                f'&cursor={cursor}')
        yield from d["results"]
        cursor = d["meta"].get("next_cursor")

class Throttle:
    def wait(self):
        dt = self.interval - (now() - self.last)
        if dt > 0: sleep(dt)
        self.last = now()
```

- Each phrase searched in titles and abstracts; results merged on the source id.
- Requests spaced by $t_{k+1} - t_k \geq \Delta$.
- On HTTP 429: back-off $w_j = \min(60j,\ 900)$ s.
- Reference lists resolved into citations (for co-citation).

Backup: full text and de-duplication

Full text

- PDFs (PyMuPDF) and web pages (trafilatura).
- Split into introduction / conclusion / section headings.

```
def key(r):
    if r.source_type == "academic":
        if r.doi: return "doi:" + r.doi.lower()
        return f"t:{norm(r.title)[:120]}:{r.year}"
    if r.source_type == "speech":
        return f"s:{r.date}:{norm(r.title)[:50]}"
    return "u:" + r.url
```

Merge: most precise date, longest abstract, union of keywords.

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